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“DESIGN AND IMPLEMENTATION OF BATTERY OPTIMIZATION SYSTEM USING SOC ESTIMATION AND PASSIVE CELL BALANCING”

Mini Project 2B

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We declare that the Mini Project report titled “**Design and Implementation of Battery Optimization System using SOC Estimation and Passive Cell Balancing**” is originally written by us and presenting our own idea. We have adhered to the Academic honesty and integrity in the preparation of the report. This Mini project report is not submitted anywhere else for any type of degree or award.

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ABSTRACT

A Battery Management System (BMS) is a vital component in modern energy storage systems, especially for lithium-ion batteries used in electric vehicles, renewable energy systems, and portable electronic devices. It ensures the safe, reliable, and efficient operation of batteries by continuously monitoring key parameters such as cell voltage, current, temperature, and overall system status. In addition to monitoring, the BMS provides protection against abnormal conditions like overcharging, deep discharging, overcurrent, and thermal runaway, thereby enhancing battery safety and extending its operational life.

One of the most important functions of a BMS is the estimation of the State of Charge (SoC), which represents the remaining usable capacity of a battery. Accurate SoC estimation is essential for predicting runtime, optimizing energy usage, and preventing battery damage. However, it is a complex task due to the nonlinear behaviour of batteries and the influence of environmental and operating conditions. Common estimation techniques include coulomb counting, open-circuit voltage methods, and advanced model-based approaches such as Kalman filtering.

Another critical challenge in battery packs is cell imbalance, which occurs due to variations in manufacturing, temperature distribution, and aging characteristics of individual cells. This imbalance can lead to reduced capacity, decreased efficiency, and potential safety hazards. Passive cell balancing is a commonly used technique to mitigate this issue, where excess energy from higher-charged cells is dissipated as heat through resistive elements to achieve uniform charge distribution. While passive balancing is simple, reliable, and cost-effective, it suffers from energy loss and slower balancing speed compared to active methods.

This report provides an in-depth overview of Battery Management Systems, the significance and challenges of SoC estimation, and the role of passive balancing in maintaining battery performance, safety, and longevity in various applications.

INTRODUCTION

A Battery Management System (BMS) is an essential electronic system used to monitor, control, and protect battery packs, especially in lithium-ion batteries. With the increasing use of battery-powered systems in applications such as electric vehicles, renewable energy storage, and portable electronic devices, the need for an efficient and reliable BMS has become very important. A BMS ensures safe operation by continuously checking key parameters like voltage, current, and overall battery condition.

In this project, a Battery Management System is designed for a 16V battery pack using components such as ESP32, ADS1115, INA219, MOSFETs (IRLZ44N), and resistors. The ESP32 acts as the main controller or brain of the system, which collects data from sensors and processes it to make decisions. The ADS1115 is used for high-precision voltage measurement of individual cells, while the INA219 measures the current flowing through the battery. These sensors provide accurate real-time data to the controller.

The system also estimates the State of Charge (SOC), which indicates how much battery capacity is remaining. This helps the user understand the battery level and usage. In addition, passive cell balancing is implemented using MOSFETs and resistors. If one cell has higher voltage than others, the system discharges the extra energy through a resistor to maintain equal voltage across all cells. This balancing improves battery performance and increases its lifespan.

Safety is one of the most important aspects of this system. The BMS protects the battery from over-voltage, under-voltage, and overcurrent conditions. If any abnormal condition is detected, the ESP32 controls the MOSFETs to disconnect the battery or limit the current flow, preventing damage and ensuring safe operation.

Overall, this project provides a simple, cost-effective, and efficient solution for battery monitoring and protection. It demonstrates the basic working principle of modern Battery Management Systems used in real-world applications, making it suitable for educational purposes as well as practical implementation.

OBJECTIVE

The primary objective of this project is to design and implement a Battery Optimization System for a 16V battery pack using State of Charge (SOC) estimation and passive cell balancing techniques. The system aims to improve battery performance, safety, and lifespan by continuously monitoring and controlling key battery parameters.

One of the main objectives is to accurately measure the voltage of individual cells and the overall current flowing through the battery using high-precision sensors such as ADS1115 and INA219. This data is processed by the ESP32 controller to estimate the State of Charge (SOC), which indicates the remaining battery capacity and helps in efficient energy utilization.

Another important objective is to implement passive cell balancing to maintain uniform voltage across all cells in the battery pack. By using MOSFETs and resistors, excess energy from higher voltage cells is dissipated, ensuring that all cells operate at equal levels. This helps in preventing cell imbalance, which can reduce battery efficiency and lifespan.

The project also focuses on enhancing battery safety by detecting abnormal conditions such as over-voltage, under-voltage, and overcurrent. The ESP32 controller takes appropriate actions, such as controlling MOSFETs to limit or disconnect the battery, thereby protecting the system from damage.

Additionally, the objective is to develop a cost-effective and efficient system that demonstrates the practical implementation of battery optimization techniques used in modern Battery Management Systems. The project provides a clear understanding of monitoring, control, and protection mechanisms, making it suitable for both academic learning and real-world applications.

LITERATURE SURVEY

The Battery Management System (BMS) is the most important part of an Electric Vehicle (EV). In the past, battery systems were very basic, but now, with advancements in technology, smart microcontrollers are used to monitor individual battery cells. This survey explains how different researchers have used modern technology to make batteries safer and more reliable.

The Use of Smart AI and Cloud Monitoring (2022)

In 2022, researchers studied how to use Artificial Intelligence (AI) and the Internet of Things (IoT) to manage batteries. They used an ESP32 microcontroller to measure battery voltage and temperature. This data was sent over Wi-Fi to the ThingSpeak platform. The study highlighted the importance of using AI to analyze data and predict possible overheating or battery failure.

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The Concept of a "Digital Twin" for Total Safety (2023)

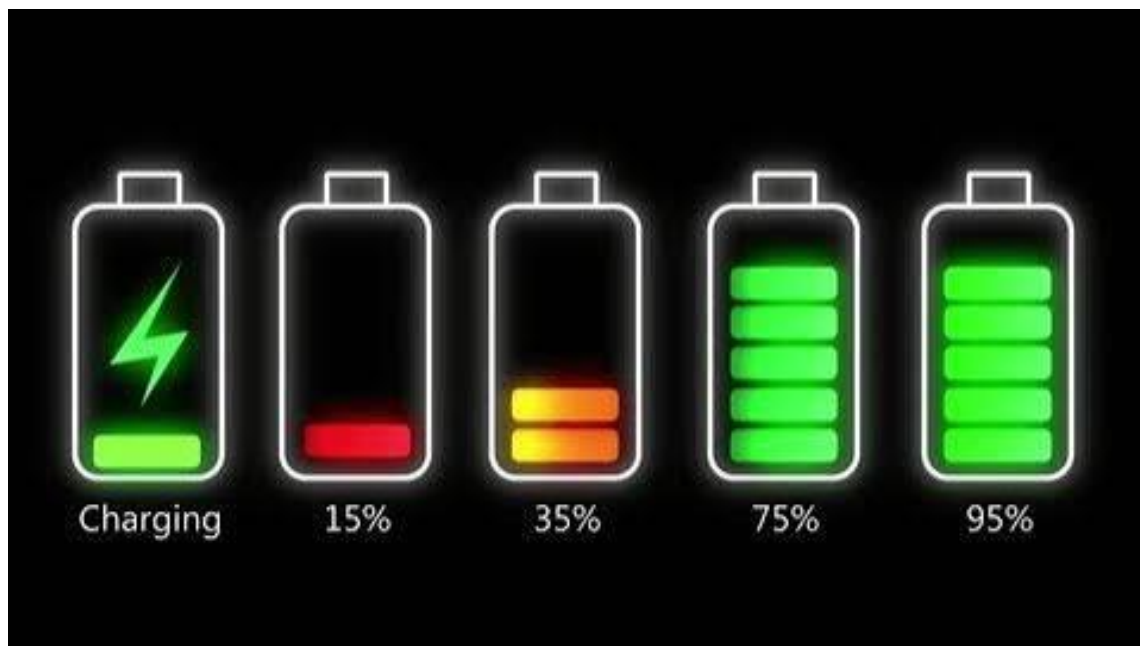
Also in 2023, a very advanced study was published about "Digital Twins." A Digital Twin is like having a "ghost" or a virtual copy of your battery inside a computer. The researchers explained that if we have very accurate sensors—like the 16-bit ADS1115 you are using—we can create a perfect digital replica. By testing the "virtual" battery under different conditions, we can see exactly how the "real" battery will behave over several years. This study is important because it proves that precision (accuracy) is more important than anything else when it depends on battery life.

Summary and Research Gap

After studying these works, I noticed a common problem: most systems either focus only on the "Internet/Cloud" side or they use low-quality sensors built into the microcontroller. My project fills this gap. Instead of relying on low-accuracy sensors, I am using a dedicated 16-bit ADC (ADS1115) for high precision. I have also added a local OLED display, so even if there is no Wi-Fi or Internet, the user can still see the battery health and stay safe.

Battery Optimization System

A Battery Optimization System is an intelligent control system designed to improve the efficiency, performance, safety, and lifespan of rechargeable batteries used in modern electronic devices and energy systems. It continuously monitors the operating conditions of a battery and makes real-time adjustments to ensure optimal energy usage and prevent damage caused by overcharging, deep discharging, or overheating. With the growing use of batteries in applications such as smartphones, laptops, electric vehicles, and renewable energy storage, maintaining battery health has become essential. The system achieves this by analyzing various electrical and thermal parameters and applying control strategies such as load management, charging regulation, and cell balancing. In battery packs consisting of multiple cells, differences in charging levels can occur due to variations in manufacturing and usage; therefore, techniques like passive cell balancing are used to equalize the charge by dissipating excess energy, ensuring uniform performance across all cells. Advanced systems may also use active balancing methods to redistribute energy more efficiently. Overall, a Battery Optimization System enhances reliability, extends battery life, and ensures safe operation under different working conditions.



Key Parameters Monitored in Battery Optimization System

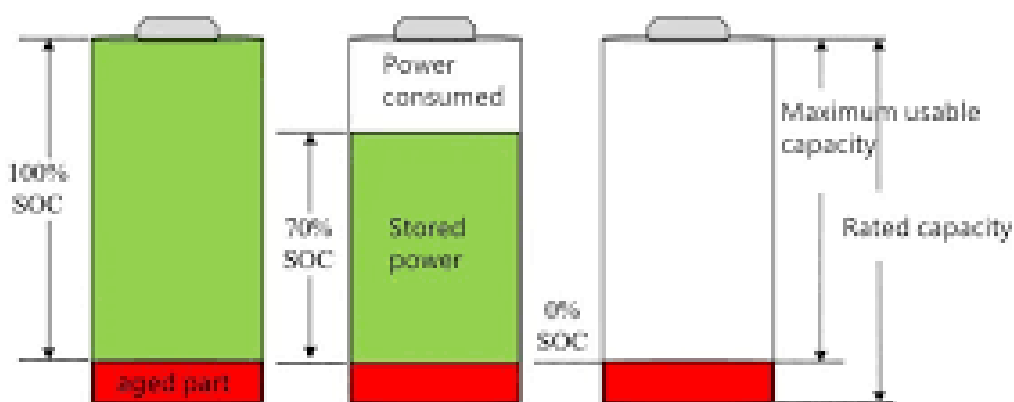
- State of Charge (SoC): Indicates the current charge level of the battery as a percentage of its total capacity.
- Passive Cell Balancing: Equalizes charge by dissipating excess energy from higher-charged cells.
- Active Cell Balancing: Redistributes energy between cells to improve efficiency and performance.
- State of Health (SoH): Represents the overall condition and aging level of the battery.
- State of Power (SoP): Defines the maximum power the battery can safely deliver at a given time.
- Voltage: Measures the electrical potential of the battery cells to ensure safe operating limits.
- Current: Monitors the flow of charge during charging and discharging cycles.
- Temperature: Tracks battery temperature to prevent overheating and thermal damage.
- Internal Resistance: Indicates energy losses within the battery and helps assess efficiency.
- Charge/Discharge Cycles: Counts the number of complete cycles to estimate battery lifespan.

State of Charge (SoC)

State of Charge (SoC) is a crucial parameter in a Battery Optimization System that represents the available capacity of a battery relative to its maximum capacity. It is usually expressed as a percentage, where 0% indicates a fully discharged battery and 100% represents a fully charged battery. SoC provides a clear indication of how much energy is remaining in the battery and helps users and systems estimate how long the battery can continue to operate before requiring recharging. Accurate SoC estimation is essential for efficient energy management, especially in applications like electric vehicles and portable electronic devices.

The determination of SoC is not straightforward, as it cannot be measured directly. Instead, it is estimated using various methods such as voltage-based estimation, coulomb counting (measuring the current flowing in and out of the battery), and advanced algorithms that combine multiple parameters like temperature, current, and battery characteristics. Modern systems often use intelligent techniques to improve accuracy under different operating conditions.

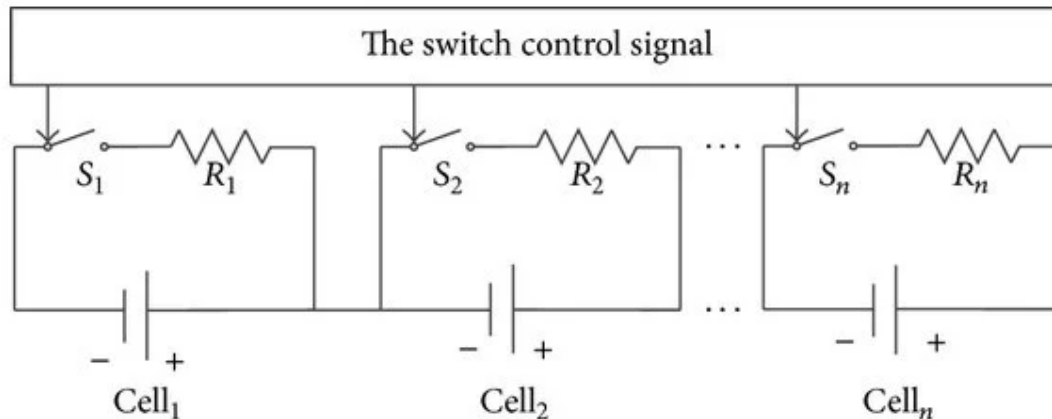
Maintaining an optimal SoC range is important for battery health and longevity. Operating a battery at extremely low or high SoC levels for extended periods can lead to degradation and reduced lifespan. Therefore, battery optimization systems use SoC data to control charging and discharging processes, prevent deep discharge or overcharging, and ensure safe and efficient operation of the battery.



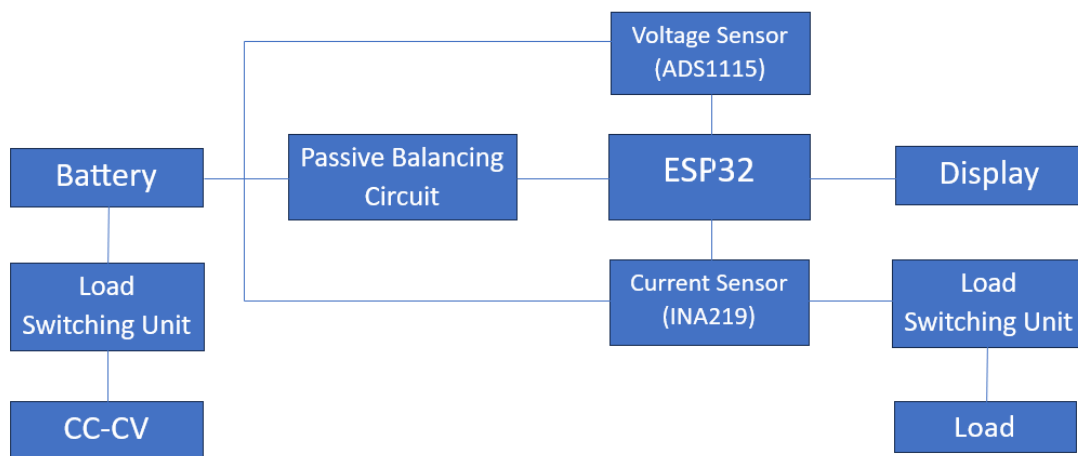
Passive Cell Balancing

Passive balancing serves as the primary mechanism for maintaining cell equilibrium within a battery pack by selectively dissipating excess energy from cells with a higher State of Charge (SoC). This process is governed by the optimization of several critical parameters to ensure pack longevity and safety. The fundamental control variable is the Voltage Threshold, which dictates the precise point at which the bypass circuit is engaged. If this threshold is set too conservatively, energy is wasted prematurely; conversely, a threshold that is too high risks cell overvoltage during the final stages of charging. To mitigate the inherent energy loss of this method, most Battery Management Systems (BMS) employ a Top Balancing strategy, optimizing the algorithm to trigger only when cells reach the upper "knee" of the voltage curve where SoC discrepancies are most measurable and impactful.

The rate of equalization is further optimized through the Balancing Current, which is inversely proportional to the value of the shunt resistor. While a higher current allows for faster convergence of cell voltages, it introduces a significant thermal trade-off. The dissipation of energy as heat necessitates a pulsed Duty Cycle optimization to prevent the BMS hardware from exceeding its thermal limits. Consequently, an effective passive balancing implementation must balance the speed of equalization against the cooling capabilities of the pack. By minimizing the SoC Deviation across the string, the system ensures that the total usable capacity of the battery is not bottlenecked by a single outlier cell, thereby maximizing the operational efficiency of the entire energy storage system.



Block Diagram:



1. Battery Pack: The battery pack consists of four lithium-ion cells connected in series (4S configuration), which increases the overall voltage while maintaining the same capacity as a single cell. Each junction between the cells provides voltage tapping points (B1, B2, B3, and B+), which are used for monitoring individual cell voltages. The battery pack acts as the main energy source of the system and supplies power to both the load and the electronic control circuit. Accurate monitoring of each cell is essential to ensure safe operation, prevent overcharging or deep discharge, and maintain balanced performance across all cells.

2. Voltage Sensor (ADS1115): The ADS1115 is a high-resolution 16-bit analog-to-digital converter used to measure battery voltages precisely. Since the battery voltages are higher than the safe input range of the ADC, they are first passed through voltage divider circuits to scale them down. The ADS1115 reads cumulative voltages from different battery taps and sends the digital data to the ESP32 via I2C communication. The ESP32 then calculates individual cell voltages by subtracting consecutive readings. This allows accurate monitoring of each cell's condition and helps in detecting imbalance or abnormal voltage levels.

3. Current Sensor (INA219): The INA219 is a current sensing module connected in series with the battery and load. It measures the current flowing through the circuit by detecting the voltage drop across an internal shunt resistor. The measured current is communicated to the ESP32 using the I2C interface. This current data is essential for calculating the State of Charge (SOC) using the coulomb counting method and for monitoring load conditions. It also helps in

identifying abnormal current flow, which can indicate faults or overload conditions.

4. ESP32 Microcontroller: The ESP32 serves as the central control unit of the system. It receives voltage data from the ADS1115 and current data from the INA219, processes this information, and calculates important battery parameters such as individual cell voltages and SOC. The ESP32 also makes control decisions based on these parameters, such as activating the balancing circuit or disconnecting the load under unsafe conditions. Additionally, it communicates with the OLED display to present real-time information to the user. Its high processing capability and built-in communication interfaces make it suitable for advanced battery management applications.

5. Passive Balancing Circuit: The passive balancing circuit consists of MOSFETs and resistors connected across each cell. The ESP32 continuously compares the voltages of all cells, and if any cell voltage is higher than the others by a predefined threshold, it activates the corresponding MOSFET. This allows current to flow through the resistor, dissipating excess energy in the form of heat. As a result, the voltage of the higher cell decreases, bringing it in line with the other cells. This process ensures uniform charge distribution, prevents overcharging of individual cells, and increases the overall lifespan of the battery pack.

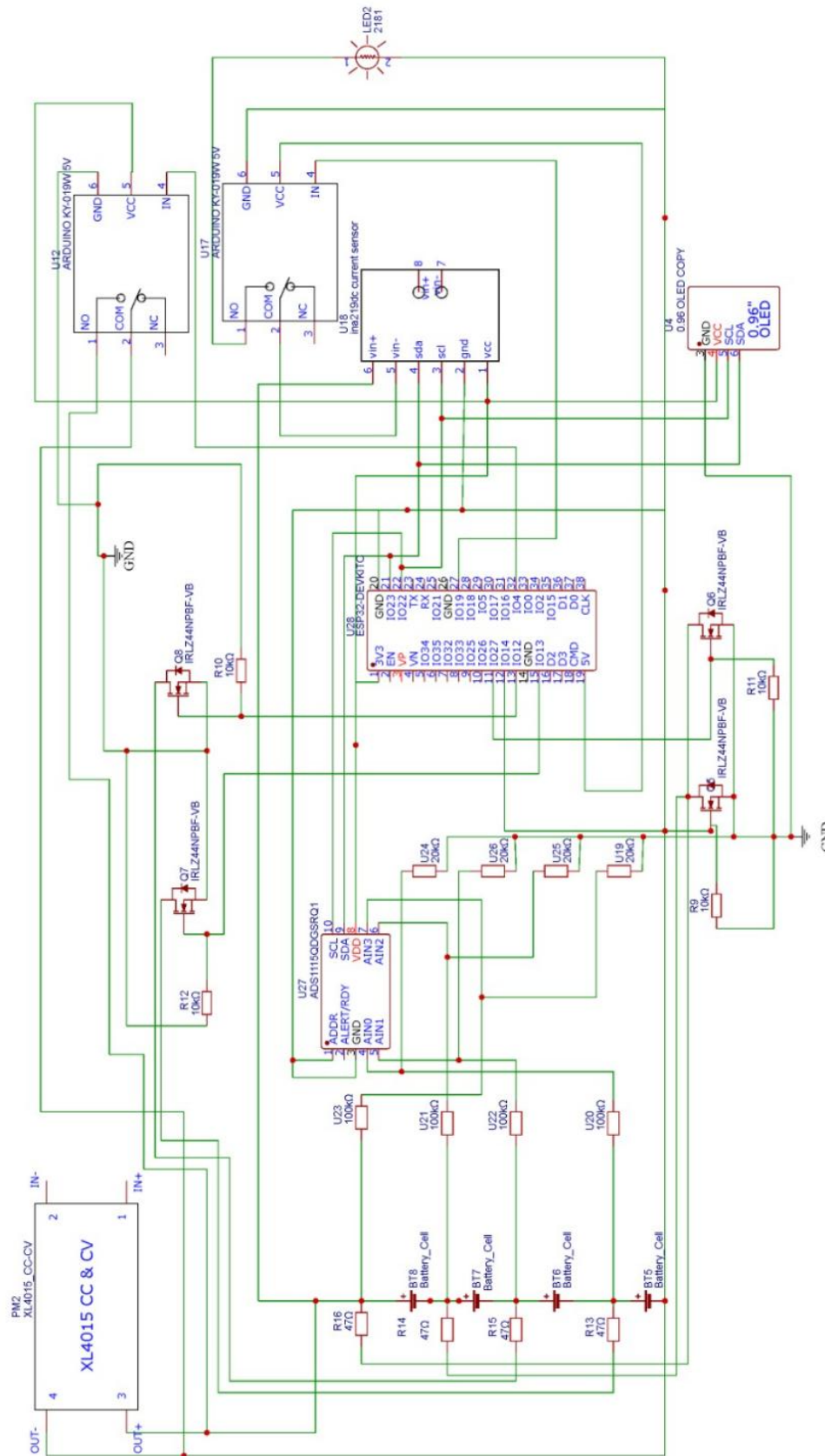
6. Load Switching Unit (Relay Module): The load switching unit is implemented using a relay controlled by the ESP32. It acts as a protective element by connecting or disconnecting the load from the battery based on system conditions. If any cell voltage drops below a safe threshold (undervoltage condition), the ESP32 deactivates the relay, thereby disconnecting the load and preventing deep discharge. This helps protect the battery from damage and ensures safe operation. The relay also allows controlled delivery of power to the load during normal conditions.

7. Load: The load represents any external device powered by the battery, such as a bulb or other electrical equipment. It receives power from the battery through the relay and current sensor. The behavior of the load is directly influenced by the control decisions of the ESP32. During normal operation, the load is powered continuously, but under fault conditions such as undervoltage, it is disconnected to protect the battery. This ensures efficient and safe utilization of stored energy.

8. Charging Unit (CC-CV Charger): The charging unit is based on the constant current–constant voltage (CC-CV) principle, which is the standard method for charging lithium-ion batteries. Initially, the charger supplies a constant current to charge the battery rapidly. As the battery voltage approaches its maximum limit (16.8V for a 4S pack), the charger switches to constant voltage mode, where the voltage is maintained and the charging current gradually decreases. This ensures safe and efficient charging, prevents overcharging, and maintains battery health.

9. Display (OLED): The OLED display provides real-time visualization of important battery parameters. It is connected to the ESP32 via I2C communication and displays information such as individual cell voltages, overall SOC, and load status. This allows the user to monitor the system continuously and understand the battery condition at any moment. The display enhances user interaction and makes the system more informative and user-friendly.

Circuit Diagram :-



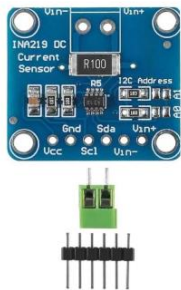
The proposed system is a microcontroller-based Battery Management System (BMS) designed to monitor, control, and protect a 4-cell lithium-ion battery pack. When the system is powered on, the ESP32 initializes all connected modules including the ADS1115 voltage sensor, INA219 current sensor, OLED display, relay, and balancing MOSFETs. The battery pack consists of four cells connected in series, and voltage taps are taken from each cell junction. These voltages are passed through a resistor-based voltage divider network to reduce them to a safe level, after which they are measured by the ADS1115 analog-to-digital converter. The ESP32 processes these readings and calculates the individual cell voltages by subtracting consecutive tap voltages.

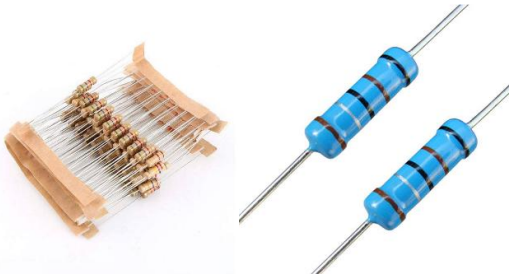
At the same time, the INA219 current sensor measures the current flowing through the load by detecting the voltage drop across its internal shunt resistor. This current data, along with time, is used for coulomb counting to estimate the State of Charge (SOC) of the battery. Additionally, a voltage-based SOC estimation is performed using the measured cell voltages. These two methods are combined to form a hybrid SOC calculation, which improves accuracy and reduces long-term drift.

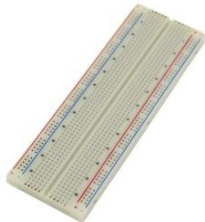
The ESP32 continuously compares the voltages of all cells to detect imbalance. If any cell has a higher voltage than others, the corresponding MOSFET in the passive balancing circuit is activated, allowing excess energy to dissipate through a resistor. This process ensures that all cells maintain equal voltage levels, thereby enhancing battery life and safety. For protection, the system checks for undervoltage conditions, and if any cell voltage drops below a predefined threshold (typically 3.0V), the ESP32 turns off the relay to disconnect the load, preventing deep discharge and damage to the battery.

The battery is charged using a CC-CV (constant current–constant voltage) charger, which first supplies a constant current and then maintains a constant voltage as the battery approaches full charge. Throughout operation, the OLED display provides real-time information such as individual cell voltages, SOC, and load status, enabling effective monitoring. The entire system operates in a continuous loop, ensuring real-time monitoring, protection, balancing, and control of the battery pack.

LIST OF COMPONENTS

<u>SR NO</u>	<u>COMPONENTS</u>	<u>SYMBOL</u>
<u>1.</u>	ESP32 Microcontroller It runs your code, takes voltage data from sensors, calculates the State of Charge (SOC), and controls the switches (Relays and MOSFETs).	
<u>2.</u>	Li-ion Batteries These store the electricity. In a 4S (4-Series) configuration, they combine to provide a nominal voltage of 14.8V (up to 16.8V max). 18650	
<u>3.</u>	Battery Holder It holds the cells securely and provides the wiring points for the Series (4S) connection.	
<u>4.</u>	Current Sensor It measures how many Amperes are entering (charging) or leaving (discharging) the battery. This data is needed to estimate the SOC (State of Charge). INA219	

<u>5.</u>	ADC Module It converts the analog battery voltage into digital data with 16-bit resolution. This allows the ESP32 to "see" tiny changes in voltage for better balancing. ADS1115	
<u>6.</u>	MOSFET These are Logic-Level MOSFETs that act as electronic switches. When a cell's voltage is too high, the ESP32 turns the MOSFET ON to start the balancing process. IRLZ44N	
<u>7.</u>	Resistor When a MOSFET is on, this resistor burns off the "extra" voltage from a full cell as a tiny amount of heat, allowing weaker cells to catch up. (47Ω, 2W), 100kΩ, 20kΩ, 10kΩ	
<u>8.</u>	Li-ion Charger It provides a Constant Current - Constant Voltage (CC-CV) charge to refill the 4S battery pack. 16.8V 2A	
<u>9.</u>	OLED Display It communicates via I2C to show the live status of the battery pack, including cell voltages, percentage, and current. 0.96"	

<u>10.</u>	Relay Modules One relay cuts off the Charger if a cell is over-charged. The second relay cuts off the Bulb if the battery is too low (Under-voltage). 5V	
<u>11.</u>	DC Bulb It acts as the device being powered. It creates the "Current Draw" so the INA219 and SOC estimation can be tested. 16V-20V DC	
<u>12.</u>	Breadboard It allows you to connect all components without soldering, making it easy to test and change the circuit.	
<u>13.</u>	Jumper Wires These provide the electrical paths for signals and power to move between the ESP32, sensors, and batteries.	

FUTURE SCOPE

The proposed battery optimization system using State of Charge (SOC) estimation and passive cell balancing can be further enhanced in several ways:

- **Integration with AI/ML algorithms**
Future systems can use machine learning models to improve SOC accuracy under varying temperature, aging, and load conditions.
- **Transition to Active Cell Balancing**
Passive balancing wastes excess energy as heat. Replacing it with active balancing can significantly improve efficiency by redistributing charge between cells.
- **Cloud-Based Battery Monitoring**
Integration with IoT platforms can enable real-time remote monitoring, predictive maintenance, and data analytics.
- **State of Health (SOH) Estimation**
Adding SOH estimation will help predict battery lifespan and degradation trends more accurately.
- **Thermal Management Integration**
Future systems can include temperature-aware optimization to prevent overheating and enhance safety.
- **Scalability for Large Battery Packs**
The system can be extended for high-capacity applications such as grid storage and electric vehicles.
- **Fast Charging Optimization**
Intelligent algorithms can optimize charging profiles to reduce charging time while maintaining battery safety.

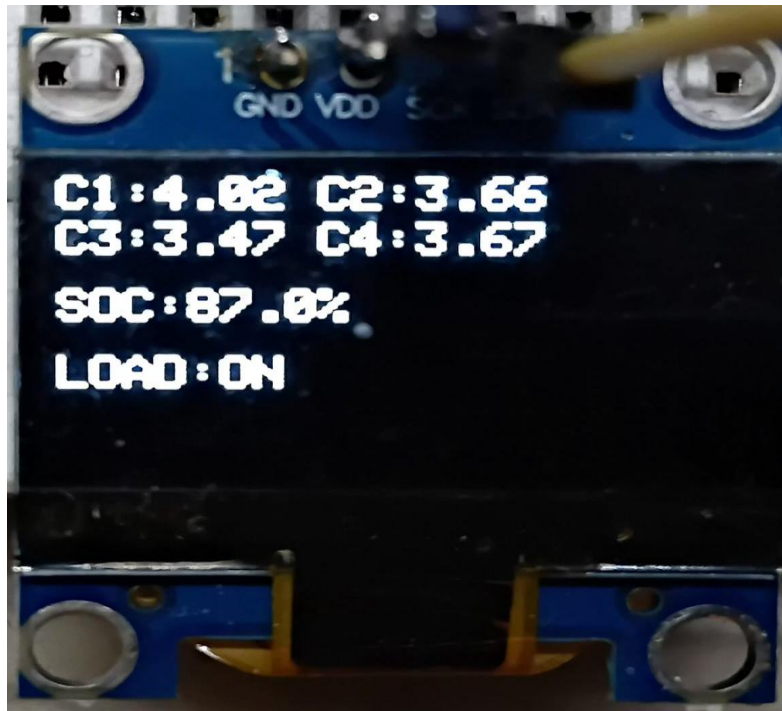
APPLICATIONS

- Electric Vehicles (EVs) : Widely used in EV battery management systems for efficient and safe operation.
- Renewable Energy Storage Systems : Used in solar and wind energy storage to manage battery banks effectively.
- Uninterruptible Power Supplies (UPS) : Ensures reliable backup power by maintaining battery health.
- Portable Electronics : Applied in devices like laptops, smartphones, and power banks.
- Electric Bikes and Scooters : Helps maintain performance and battery longevity in light EVs.
- Grid Energy Storage : Used in large-scale energy storage systems for load balancing and backup.
- Aerospace and Defence Systems : Critical in applications requiring high reliability and safety.

ADVANTAGES

- Improved Battery Life : Proper SOC estimation and balancing prevent overcharging and deep discharge, increasing lifespan.
- Enhanced Safety : Reduces risks like thermal runaway, overvoltage, and cell imbalance.
- Cost Efficiency : Extends battery replacement intervals, reducing overall operational cost.
- Better Energy Utilization : Ensures uniform charge distribution across cells, improving efficiency.
- Simple and Low-Cost Implementation : Passive balancing circuits are easy to design and require fewer components.
- Reliable Performance : Maintains consistent output by avoiding weak-cell domination in battery packs.

RESULT



CONCLUSION

This project successfully presented the design and implementation of a battery optimization system using State of Charge (SOC) estimation and passive cell balancing. The developed system was able to monitor battery charge levels accurately and reduce cell imbalance within the battery pack. By employing passive balancing, excess charge in higher-voltage cells was dissipated to maintain voltage uniformity, improving battery performance, safety, and operational life.

The results demonstrated that the proposed system enhances energy utilization, prevents overcharging and deep discharge, and improves the reliability of battery management systems. This project also highlighted the practical importance of battery optimization in applications such as electric vehicles, renewable energy storage, and industrial power systems. Overall, the developed system provides a cost-effective and efficient solution for battery monitoring and optimization, with future scope for implementing active balancing techniques and advanced SOC estimation methods for further improvement.

REFERENCES

- Title: Artificial Intelligence and IoT based Smart Battery Management System for Electric Vehicle
Author: S. Umathe and R. Hiware; G.H. Raison College of Engineering, Nagpur; 2022
This paper shows how to build a smart battery monitor using an ESP32 and a current sensor. The main goal was to send battery data (like voltage and heat) to a mobile app using Wi-Fi. This helps users get instant alerts if the battery is in danger, making electric vehicles much safer and cheaper to maintain.
- Title: IoT Based Battery Monitoring System for Electric Vehicle using ESP32
Author: R. Usha Rani, G. Divya, et al.; VNR Vignana Jyothi Institute of Engineering and Technology; 2023
This work acts as a guidebook for connecting battery hardware to the cloud. It explains how the ESP32 can collect raw data—like how much charge is left—and send it to the Arduino IoT Cloud. This is very useful for finding faults or errors in the battery pack from a distance without needing to check it physically.
- Title: Digital twin of electric vehicle battery systems: Comprehensive review of use cases and platforms
Author: F. Naseri, S. Gil, et al.; Aarhus University, Denmark / Stanford University; 2023
This research talks about a "Digital Twin," which is like creating a computer-based twin of a real battery. By using very accurate sensors to feed data into a computer model, the system can predict the battery's future health. This is the most advanced way to ensure a battery never fails unexpectedly.